



C#

Level 1

Methods



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Definition

Method

-  A method is a block of reusable code that runs when it is called.
-  It can perform an action, receive parameters, and return a value.

Explanation

Basic Syntax

```
static void MyMethod()  
{  
  // code  
}
```

 static → allows calling the method without creating an object.

 void → the method does not return a value.

 MyMethod → method name.

 () → contains parameters, if any.

 {} → contains the method's code.

Calling a Method

 Defining a method does not execute it.

 The method runs when it is called using its name followed by ().

💡 Example

```
internal class Program
{
    static void PrintMyName()
    {
        Console.WriteLine("Enter your Name: ");
        string myName = Console.ReadLine();
        Console.WriteLine("Welcome " + myName);
    }
    static void Main(string[] args)
    {
        PrintMyName();

        Console.ReadKey();
    }
}
```

➡ Result

```
Enter your Name:
Sara
Welcome Sara
```

Note

- ◆ Methods must be defined inside a class or another type.
- ◆ Main() is also a method and is the entry point of a console application.



Comparison



static method

- ✓ Called directly through the class.
- ✓ Belongs to the class and can be called directly without creating an object.



non-static method

- ✓ Called through an object.
- ✓ Normally requires an object to be created before calling it.

✦✦ Summary

- ✓ Methods contain reusable code that runs when called.
- ✓ They can receive parameters and return values.
- ✓ void means no value is returned.
- ✓ static allows calling a method without creating an object.

Thank you 😊